Class #3 - Row Reduction Algorithm

1. Illustration:

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix}$$

   Start with the 1st nonzero column, and work out by each column.
   (*Pivot* (or *pivot element*): the element of a matrix which is selected first by an algorithm to do certain calculations.)

2. Def: A *pivot position* in a matrix is the position of a pivot in its echelon form, and a *pivot column* of a matrix is the column which has a pivot.

   Example: Pivot positions of

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix}$$

   are __________________________.

   Pivot columns are __________________________.

3. Example: The following is an augmented matrix of a linear system. Reduce the matrix to its echelon form. Is it consistent?

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$$

4. A *variable* corresponding to a pivot column is called a *basic variable*; A variable corresponding to a nonpivot column is called a *free variable*.

5. Example: Consider the following augmented matrices in its echelon form. Discuss solution set for each where $\square$ means pivot and '*' means any real number:

$$\begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & 0 & \square \end{bmatrix} \qquad \begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & \square & * \end{bmatrix} \qquad \begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

6. Example: Suppose a 3×5 coefficient matrix for a system has three pivot columns. Is the system consistent? Why or why not?